REVIEW

Assessing Sub-Regulatory PFAS Exposure in Drinking Water as a Predictor of Allergic Disease in Pediatric Populations

Baura Galym[1], Daneen Chaudhry[2], Hashir Adnan[3], Niginabonu Mehmonova[4], Roberto Gonzalez[5]

[1]Binom School named after Alikhan Bokeikhan, Nura District, Astana, Kazakhstan
[2]Beaconhouse Karakoram Campus, Islamabad, Pakistan
[3]DuPont Manual High School, Kentucky, USA
[4]School No. 34, Bukhara, Uzbekistan
[5]Carl Wunsche Sr. High School, Texas, USA

Abstract

Per- and polyfluoroalkyl substances (PFAS) are synthetic chemicals containing very strong carbon-fluorine bonds. Therefore, they are resistant to the natural processes of degradation. They are found in every-day items of use, such as food packaging, grease and water. Consequently, they make their way inside the human body, making individuals highly susceptible to allergies and hypersensitivity because they alter the usual immune cell functions. As a result, environmental protection agencies across multiple countries have tightened regulations regarding PFAS concentrations in water. However, these limits are only developed after assessing the effects of high-dose exposure on other functions of the human body, leaving the effects of sub-regulatory PFAS concentrations in drinking water on the immune systems of children poorly investigated. To address the steady global rise in pediatric allergies, this study chose a systematic review and a comparative analysis to examine whether sub-regulatory PFAS exposure was associated with reported allergic disease prevalence. To do this, we conducted a thorough research of publicly available databases,including PubMed and ScienceDirect, collecting quantitative data on PFAS concentrations in drinking water, alongside allergic disease prevalence in the same region. This led to the development of 2 datasets–one PFAS concentration data set and one allergic disease prevalence dataset–geographically matched across 11 locations, including Japan, Canada, Norway, the United States, and United Kingdom, with a focus on regions with sub-regulatory PFAS concentration data. Concurrently, reported allergic disease and asthma prevalence varied widely, ranging from 5% in Norway to 23.9% in Shanghai, China. The data was visualised through comparative graphs, showing a general pattern for higher allergic diseases among children in regions where sub-regulatory PFAS compounds were present. However, this observed pattern was not uniform across all matched geographical locations. Furthermore, no formal statistical correlation coefficients were calculated. Henceforth, the results suggest potential association–rather than definitive causation–between sub-regulatory waterborne PFAS and pediatric allergic susceptibility, confirming that existing safety standards may be insufficient to fully safeguard child health.